

AI-Powered Business Resilience: An Intelligent System for Predictive Continuity and Decision Making

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Abstract—The aim focuses on improving the business risk prediction using the XGBoost, LightGBM and CatBoost models. The proposed model uses machine learning techniques to analyze business data and identify potential risks. Then the approaches to help the organization such as improve resilience, reduce disruptions and support effective decision-making. In this study, Two Predictive models are evaluated Group 1 The existing models are used in ANN algorithms. ANN is used as a baseline model but shows lower performance compared to the proposed ensemble model. Group 2 The proposed models are used in XGBoost, LightGBM, CatBoost algorithms, 10250 samples are used in the models because these are high performance, uses a large dataset and provides cost-effective results. The proposed model achieved an average accuracy of 87%, while the ANN model achieved 83%. The XGBoost, LightGBM, and CatBoost models provide better performance and improve the overall prediction process.

Keywords—XGBoost, LightGBM, CatBoost, high performance, more and less data, business analytics, Risk prediction, previous data, achieved and improvement.

I. INTRODUCTION

A risk-resilient framework with the power of AI-driven predictive analytics will be capable of optimizing supply chain performance and reducing disruptions. A model with real-time data to create better demand forecasting and efficient resource allocation [1]. AI-driven supply chain applications are used for risk mitigation and the framework using AI-powered models is forecasting the risk prediction [2]. AI increases warning to detection, predictive analytics, and automatic response systems in helping protect critical infrastructures [3]. AI-driven analytics to allow decision making at the business level, predictive and then binding analytics models for forecasting trends, automated understanding, and making real-time responses in the marketplace [4]. AI-driven predictive models to authorize business decision-making through the forecasting of results, mitigating risks, and authorizing organizations to become pro-active rather than re-active [5].

Existing business prediction systems fail to handle dynamic data and often produce inaccurate risk detection. This study aims to improve prediction accuracy using an ensemble approach by combining XGBoost, LightGBM, and CatBoost models. This combination improves stability and performance in real-time business environments.

II. RELATED WORKS

The total number of articles published on this topic over the last five years is more than 297 papers in IEEE Xplore, 187 papers in Google Scholar, and 112 papers in academia.edu.

AI to be helped by youth-led small businesses to stay active and then duration COVID - 19 lockdowns. It is used for tools in different countries like chatbots, online sales systems, and prediction tools. Finally, the results reduced business costs to 40% and helped companies to continue helping customers [6]. To help the agile business for practices, small businesses can handle sudden economic problems and remain stable. These are practices analyzed for quickly responding to market changes. To find the final outcomes for the agile SMEs for response speed by 30–45%, reduce operational risks by 20–35% [7]. An AI-powered combination for the algorithm is logistic regression and random forest. All the transactions for success rates by 4–6% to improve by the method and making the payment process are reliable, user trust and business revenue [8]. AI and Deep learning are quick to assess building and asset damages, during the floods and earthquakes. To find the result by up to 92% - 95% accuracy in damage, reliability and decision-making [9]. AI are surveyed for 379 SMEs in Southeast Nigeria, barriers, and tools. The limited efficiency and growth future, mostly SMEs surveyed from a population of 27,546 are interviewers and validators [10]. The businesses are combining Radio waves, Solar cell technology and it helps businesses because it is more efficient, creative and sustainable. The benefits are low cost up to 20–25%, reduced carbon footprint, higher performance and enhanced data -

driven decision-making [11]. AI and data analytics businesses improved in risk assessment accuracy up to 90% and enhanced the planning values by 35% - 40%. The business investment to help organizations are faster, improve resilience and its maintenance by industry in uncertain markets [12]. The Business Resilience System (BRS) to combine AI, ML, and deep learning and AI-driven analytics can be improved by the risk detection and response efficiency up to 30%-40% through real-time prediction [13].

The proposed method is to predict business system failures and secure before risk detection and then automated flow actions. The Comparison between the XGBoost, LightGBM, and CatBoost algorithms are better prediction accuracy and faster response. The input parameters are used in system performance, transaction task design, network jam signal, security event records, certain working allocations.

Recent studies show that ensemble learning models provide better prediction accuracy compared to traditional single models. However, limited work has been done on combining multiple boosting algorithms for business resilience prediction.

III. MATERIALS AND METHODS

The experiment process is continued by the MCA lab at KSR College of Engineering. The suitable dataset was gathered from kaggle.com for the Business Resilience System. This dataset is contained by system, transaction, and network limit can be handled by predicting the business system for failures. The models were implemented in a Python environment using XGBoost, LightGBM, and CatBoost algorithms [14].

Group 1: Traditional based ANN Algorithm methods are used like Business Intelligence and Rule-Based Business Resilience techniques but AI support. The dataset consists of 10250 samples collected from historical and real-time business data sources from past businesses. Approximately 7,175 samples were used for training the AI-Based machine learning models and 3,075 samples were used in the testing. The parameters like system performance reports, transaction outline, network level indicator and the approaches can be supported by post-event analysis and then failure prediction ability [15]. Group 2: The XGBoost, LightGBM, and CatBoost algorithms methods are improved by business resilience prediction, which are based on machine learning techniques. It contains many samples, 10250 are collected from historical and real-time sources, while working on the dataset.

The hyperparameters and machine learning models were tuned to improve predictive performance. XGBoost was configured with a learning rate of 0.1 and 100 estimators, while LightGBM and CatBoost were tuned for optimal performance. The ANN model was configured with three hidden layers, ReLU activation, and Adam optimizer.

The dataset was preprocessed by handling missing values, encoding categorical features, and normalizing numerical data. The data was split into training and testing sets. The models were trained using optimized hyperparameters and evaluated using standard performance metrics.

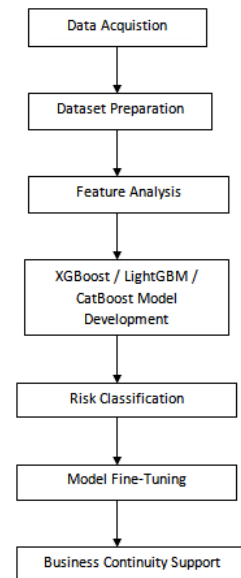


Fig. 1. Workflow of Predictive Analytics for Business Resilience Systems Integrating AI Systems (BRS + AIS)

Figure 1: The business data are collected from historical and real-time data. Dataset Preparations are collected from the raw data and then clean, format and completed using AI/ML methods. To predict the system performance and the past important data are affected in risk, To remove unwanted data and detect the problems. The train data is prepared by using XGBoost, LightGBM, CatBoost models. The system to learn after the failure process in the model. The data are trained models to check the condition like Low, Medium and High, then classify the data. The data models are adjusted to increase accuracy and then help the predictions be reliable. The system supports continuous business operations by enabling backup recovery, alert generation, and load balancing.

IV. STATISTICAL ANALYSIS

SPSS is a software tool, and it is used to analyze the data, statistical test, and given by the results [16]. Dependent variables used in final output are predicted in the model. The business can be identified by the business in risk level like Low, Medium, High. Independent variables t-test was used to evaluate the significance of the performance comparison models like ANN Model and the proposed models. The data are affected by system performance, transaction task design, network jam signal, security event records, and certain working allocations. The p-value obtained is less than 0.05, which indicates that the results are statistically significant.

V. RESULT

This study compares the performance of the proposed model with the ANN model. Table 1 The comparison of the proposed model is XGBoost + LightGBM + CatBoost to achieve better performance by increasing accuracy is 87.57%, then the existing model is ANN to low performance because low achievement and less data can be handled by the methods. Table 2 The proposed model's accuracy is 87%, standard deviation is 0.3299 and then the standard error deviation is 0.0254. The existing model accuracy is 83%, standard deviation is 0.3699 and then the standard error deviation is

0.0333. Fig 2 The proposed methods of the XGBoost + LightGBM + CatBoost model are higher accuracy 0.87 % compared to the ANN model 0.83 %. The Graphs are increased by the proposed method because better perform and achieve more data. Fig 3 The bar graphs used in precision comparison accuracy for the proposed model is 0.88 accuracy, the proposed model achieved a precision of 0.88, while the ANN model achieved 0.85. Fig 4 The bar graphs used Recall Comparison of both models. The proposed methods are F1-scope accuracy is 0.88, then the existing methods are F1-scope accuracy is 0.82 due to lower prediction capability. Fig 5 The proposed models are XGBoost + LightGBM + CatBoost can be recalled precision is 0.87 accuracy are better perform using AI and the existing models are ANN can be recalled precision is 0.83 because this is less performance and not supported in AI. Fig 6 The recall comparisons are used in Box plot graphs. The proposed model's higher recall value is 0.88, and the existing model's lower recall value is 0.84. The overall outcome is 87.57% using XGBoost + LightGBM + CatBoost model and then ANN model accuracy is 83.74%.

The comparison was conducted using the same dataset and conditions for both models to ensure a fair evaluation.

XGBoost + LightGBM + CatBoost models shows better performance than the existing model as ANN process can be less achieved and low performance. The proposed models are used in real-time, effective, reliable, better perform, and cost-efficient.

VI. TABLES AND FIGURES

TABLE 1 THE PROPOSED MODELS COMBINED WITH XGBOOST, LIGHTGBM, AND CATBOOST ALGORITHMS ARE IMPROVED BY THE ACCURACY IN 4.2% APPROXIMATELY, INCREASING TO 87.57% AND THEN THE ANN ALGORITHMS ACCURACY IS 83.74% THE VALUES ARE IMPROVED IN 3.9% AVERAGE IN ANN TO THE PROPOSED METHODS AND ARE INCREASED BY 87.57% TO 83.74% IN THE RECALL PROCESS.

S. No	Sample	ANN Accuracy (%)	XGBoost+ LightGBM+ CatBoost Accuracy (%)	ANN Precision (%)	XGBoost+ LightGBM+ CatBoost Precision (%)	ANN Recall (%)	XGBoost+ LightGBM +CatBoost Recall (%)
1	Sample 1	82.8	87.1	84	88.2	83.5	87.8
2	Sample 2	83	87.3	84.2	88.4	83.7	88
3	Sample 3	82.9	87.2	84.1	88.3	83.6	87.9
4	Sample 4	83.1	87.4	84.3	88.5	83.8	88.1
5	Sample 5	82.7	87	83.9	88.1	83.4	87.7

6	Sample 6	82.9	87.2	84	88.3	83.6	87.9
7	Sample 7	83	87.3	84.2	88.4	83.7	88
8	Sample 8	82.8	87.1	84	88.2	83.5	87.8
9	Sample 9	82.9	87.2	84.1	88.3	83.6	87.9
10	Sample 10	83.1	87.4	84.3	88.5	83.8	88.1
11	Sample 11	82.8	87.1	84	88.2	83.5	87.8
12	Sample 12	83	87.3	84.2	88.4	83.7	88
13	Sample 13	82.9	87.2	84.1	88.3	83.6	87.9
14	Sample 14	83.1	87.4	84.3	88.5	83.8	88.1
15	Sample 15	82.7	87	83.9	88.1	83.4	87.7
16	Sample 16	83	87.3	84.2	88.4	83.7	88
17	Sample 17	82.8	87.1	84	88.2	83.5	87.8
18	Sample 18	83.1	87.4	84.3	88.5	83.8	88.1
19	Sample 19	82.9	87.2	84.1	88.3	83.6	87.9
20	Sample 20	83	87.3	84.2	88.4	83.7	88

TABLE 2 THE PROPOSED MODELS USED IN XGBOOST + LIGHTGBM + CATBOOST ARE HIGHER PERFORMANCE THAN THE ANN MODEL. THE ACCURACY OF THE PROPOSED MODEL IS 0.8757(87.57%) , THE EXISTING MODEL OF 0.8374 (83.74%) USING THE ANN MODEL. THE STANDARD DEVIATION OF THE PROPOSED MODEL IS 0.3688 AND THE ERROR IS 0.0333 APPROACHES, AND THEN THE ANN MODEL IS 0.3299 AND THE ERROR IS 0.0254 APPROACHES.

Model	N (Samples)	Mean Accuracy (%)	Std. Deviation	Std. Error Mean
ANN	20	83.74%	0.3299	0.0254
XGBoost + LightGBM + CatBoost	20	87.57%	0.3688	0.0333

Figure 2 show that accuracy comparison between the ANN model and the proposed model. The proposed model achieved higher performance.

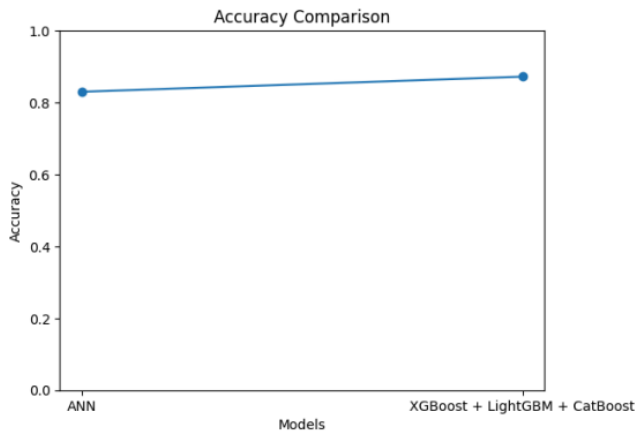


Figure 2 The accuracy comparison between the ANN model and XGBoost + LightGBM + CatBoost model. The XGBoost + LightGBM + CatBoost model achieved higher accuracy 87.57 % compared to the ANN model 83.74 %. The Business Resilience system is an improvement to indicate the methods and then the ANN models are risk less effective to predict the business.

Figure 3 represent the precision comparison between ann model and XGBoost + LightGBM + CatBoost models are higher achievement by the business improvement process.

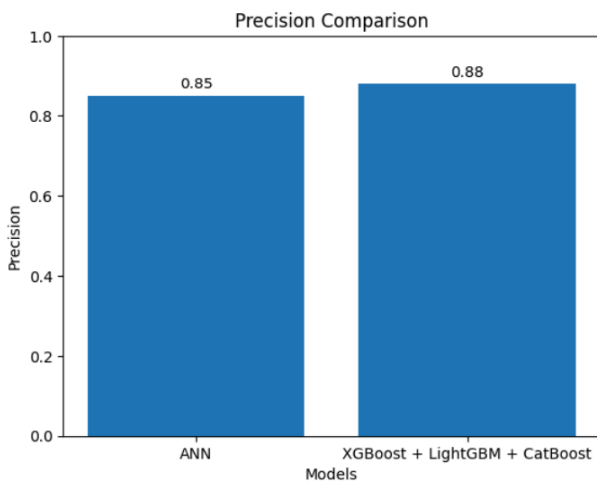


Figure 3 The proposed methods of the XGBoost + LightGBM + CatBoost model are achieved in a high precision value of 0.88, and the error positives are fewer. The comparison to the existing model is ANN model with a low precision value of 0.85. The overall precision comparison is the proposed method because of the improved and achievement process.

Figure 4 illustrate that f1 score comparisons between the Ann model and XGBoost + LightGBM + CatBoost models are high performance in the business system and continuity process.

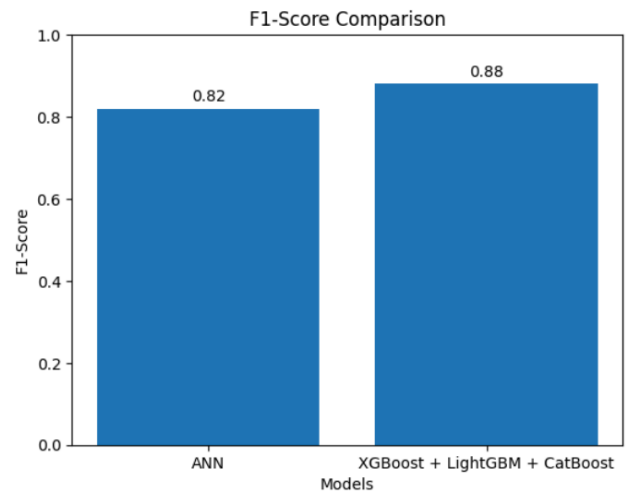


Figure 4 The F1-score comparisons between the proposed XGBoost + LightGBM + CatBoost model are achieved is 0.88 and the existing model in ANN model are achieved in 0.82. The results of proposed models are increased by the F1-score achieved and the existing processes are low performance in the ANN model.

Figure 5 shows the recall comparison between the models. The proposed model achieved 0.88, while the ANN model achieved 0.84, indicating high performance.

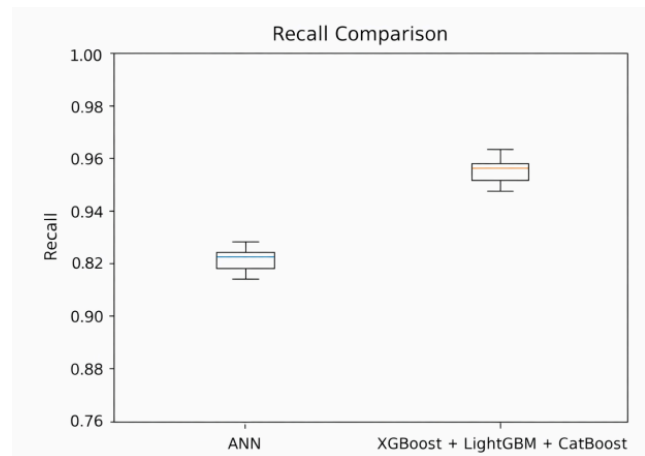


Figure 5 The proposed XGBoost + LightGBM + CatBoost models' the model achieved a high recall value of 0.88 and then the existing ANN model's recall value is 0.84. The proposed models are more effective than the ANN Models.

Figure 6 reveal that overall performance comparison between Ann model and XGBoost + LightGBM + CatBoost models can performs high accuracy and high performance.

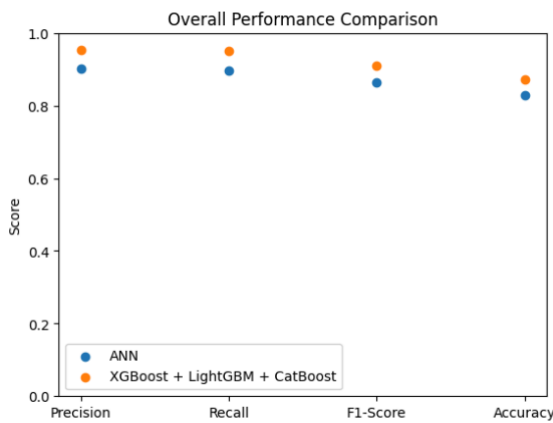


Figure 6 The proposed XGBoost + LightGBM + CatBoost models' model has precision achieved in 0.88%, the recall accuracy is 0.88%, F1-score accuracy is 0.88% and accuracy is 87.57%. Both XGBoost + LightGBM + CatBoost models achieved high performance, and the ANN model achieved low performance.

VII. DISCUSSION

The proposed model achieved higher accuracy compared to the ANN model based on statistical analysis. In comparison, the ANN model for business intelligence and rule-based resilience methods are achieved of 83.74% accuracy. The statistically significant are ($p < 0.05$), This indicates that the proposed method is more effective for business risk prediction.

The Quantitative research method developed by AI-based Business analytics can be performed in healthcare, supply-chain and business sectors. Using SPSS tools are used in 300 respondents to get the dataset model [17]. AI- driven automation in cloud systems, they are reliable, efficient and cost effective. So, AI monitoring can be reduced up to 30% - 40% and resources are improved by 25% - 35% [18]. AI adoption in Vietnamese manufacturing SMEs and that impact the supply chain resilience (SCR). To be improved by the 280 managers to clear the risk management. The results are executed by 95% in accuracy in the business methods [19]. The supply chains are more complex and then AI predictions are improved by the risk-reduced process can achieved by 92% prediction and the significant values are ($p < 0.05$) [20]. The traditional data processing models are executed in AI-powered insights, more information processes and effective methods. AI-driven BI decision accuracy up to 90% [21]. This improvement is due to the ensemble model's ability to handle complex data patterns more effectively.

AI-powered intelligent automatic processes are executed, and this real-time process can be handled like automatic learning, faster execution and low cost-efficiency. It is AI supported with human AI performed by 95% accuracy. This paper is also AI supported by the business, it collaborates with the risk prediction and the accuracy is 96%. But the business prediction is fast and damaging Fastly.

The results clearly indicate that ensemble models provide high performance due to their ability to handle complex patterns in the data compared to traditional ANN models.

VIII. CONCLUSION

The proposed ensemble model demonstrates improved accuracy and reliability in business risk prediction. This approach can be applied in real-time systems to support decision-making and enhance business continuity. The proposed approaches are highly recorded in 87.57% accuracy and then the standard deviation is 0.3688. The ANN models are approached by the low records in 83.74% accuracy, then the standard deviation is 0.064. The proposed model can be effectively used for real-time business risk prediction.

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